Geometric Origin of CP Phases from Hyperbolic Holonomy

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Abstract

We show that CP-violating phases arise naturally as holonomy invariants associated with flat $U(1)$ connections on compact hyperbolic 3-manifolds. If $M = \mathbb{H}^3/\Gamma$ admits an orientation-reversing isometry Θ, then Θ induces complex conjugation on $U(1)$ character representations of $\pi_1(M)$. Any closed geodesic whose holonomy is not ± 1 therefore provides a rephasing-invariant CP-odd observable. For flat connections these phases are topological invariants and are stable under all metric deformations preserving topological type. We formulate the argument entirely in representation-theoretic language and illustrate the mechanism using the Meyerhoff manifold (OrientableClosedCensus[1], vol $= 0.9814$, $H_1 = \mathbb{Z}/5$).

The geometric construction developed here provides the theoretical foundation for the companion results in [10, 11], where the CKM and PMNS mixing matrices are derived explicitly from the K- and N-factors of the Iwasawa decomposition of compact hyperbolic 3-manifold holonomy [10, 11].

1 Hyperbolic Setup

Let Γ be a torsion-free discrete group acting properly discontinuously and cocompactly on $\mathbb{H}^3$ by orientation-preserving isometries. Let

$$M = \mathbb{H}^3/\Gamma$$

be the resulting compact oriented hyperbolic 3-manifold.

Closed geodesics correspond to conjugacy classes $[\gamma] \subset \Gamma$ with γ hyperbolic.

[Flat $U(1)$ connection] A flat $U(1)$ connection on M is equivalently a character

$$\rho : \pi_1(M) \longrightarrow U(1).$$

The holonomy of a closed geodesic represented by γ is

$$W(\gamma) := \rho([\gamma]).$$

Because $U(1)$ is abelian, $W(\gamma)$ depends only on the free homotopy class.

2 Orientation Reversal and CP

Let $\Theta : M \to M$ be an orientation-reversing isometry. It induces an automorphism

$$\Theta_* : \pi_1(M) \to \pi_1(M).$$

[CP action on characters] Given a character ρ, define its CP-conjugate character by

$$\rho^\Theta(\gamma) = \overline{\rho(\Theta_*^{-1}\gamma)}.$$

The inverse Θ_*^{-1} appears because Θ induces a *pullback* of representations: if Θ_* is an automorphism of $\pi_1(M)$, the natural action on characters is by precomposition with Θ_*^{-1}, ensuring ρ^Θ is again a group homomorphism.

[Holonomy conjugation] Let ρ be a flat $U(1)$ character. For any $\gamma \in \pi_1(M)$,

$$W_{\rho^\Theta}(\gamma) = \overline{W_\rho(\Theta_*^{-1}\gamma)}.$$

By definition,

$$W_{\rho^\Theta}(\gamma) = \rho^\Theta(\gamma) = \overline{\rho(\Theta_*^{-1}\gamma)} = \overline{W_\rho(\Theta_*^{-1}\gamma)}.$$

[Geometric CP violation] Suppose:

1. M admits an orientation-reversing isometry Θ,
2. ρ is a flat $U(1)$ character,
3. there exists a hyperbolic element γ whose conjugacy class is fixed by Θ_* (i.e., $\Theta_*[\gamma] = [\gamma]$) and such that $W_\rho(\gamma) \neq \pm 1$.

Then $\arg W_\rho(\gamma)$ is a rephasing-invariant CP-odd observable.

Because the conjugacy class of γ is fixed, we may choose a representative so there exists $\delta \in \pi_1(M)$ with $\Theta_*^{-1}(\gamma) = \delta\gamma\delta^{-1}$. Since $U(1)$ characters are conjugation-invariant, $W_\rho(\Theta_*^{-1}\gamma) = W_\rho(\gamma)$ exactly. Applying the CP action,

$$W_\rho(\gamma) \xrightarrow{\text{CP}} W_{\rho^\Theta}(\gamma) = \overline{W_\rho(\Theta_*^{-1}\gamma)} = \overline{W_\rho(\gamma)}.$$

Hence $\arg W_\rho(\gamma) \mapsto -\arg W_\rho(\gamma)$. If $W_\rho(\gamma) \neq \pm 1$, the phase is nonzero and flips sign under CP, so it is a genuine CP-odd observable. Rephasing invariance follows from Proposition 3.

3 Rephasing Invariance

Holonomy phases are invariant under all $U(1)$ gauge transformations.

Gauge transformations act by conjugation on ρ, but since $U(1)$ is abelian, conjugation is trivial.

The term *rephasing invariance* here refers to $U(1)$ gauge transformations of the flat connection, which are trivial for an abelian structure group. This is distinct from the sector-local rephasings of the companion papers [10, 11], which remove unphysical phases from fermion field definitions. Both notions confirm that $\arg W_\rho(\gamma)$ is invariant and carries physical information.

4 Topological Stability

Let ρ be a flat character. Then $W_\rho(\gamma)$ depends only on the homotopy class of γ. It is invariant under:

1. metric deformations preserving the topological type of M,
2. continuous deformations of ρ within its connected component in the character variety.

Flatness implies holonomy depends only on the homotopy class. Metric deformations do not change $\pi_1(M)$, so the holonomy of each class remains unchanged. Continuous variations of ρ in character space vary the holonomy continuously; if they stay within the same component, the holonomy can change continuously but remains a topological invariant in the sense that its value is determined by the representation. For the purpose of CP violation, it suffices that the phase is topologically quantised (e.g., a root of unity) and does not drift under small metric perturbations.

[Discrete character variety] When $H_1(M;\mathbb{Z})$ is purely torsion—as for the Meyerhoff manifold with $H_1 = \mathbb{Z}/5$—the character variety $\mathrm{Hom}(\pi_1(M), U(1))$ consists of finitely many isolated points. For $\mathbb{Z}/5$ there are exactly five, corresponding to the fifth roots of unity. Each connected component is a single point, so no continuous deformation of ρ is possible; the holonomy phase is *topologically quantized* and cannot drift.

5 Example: The Meyerhoff Manifold

The Meyerhoff manifold M = OrientableClosedCensus[1] (SnapPy [3], vol = 0.9814) is proven arithmetic with invariant trace field $\mathbb{Q}(\sqrt{-283})$ [4]. Its first homology group satisfies

$$H_1(M;\mathbb{Z}) \cong \mathbb{Z}/5.$$

The character variety consists of five isolated points; choose ρ sending the generator to $e^{2\pi i/5}$ and let γ represent the generator of $\mathbb{Z}/5$. Complex conjugation in $\mathbb{Q}(\sqrt{-283})$ induces an orientation-reversing isometry acting by inversion $k \mapsto -k$ on $H_1(M;\mathbb{Z}) = \mathbb{Z}/5$, mapping χ to its complex conjugate [5, 4]. The conjugacy class of γ is fixed, so Theorem 2.3 applies. Therefore

$$W_\rho(\gamma) = e^{2\pi i/5},$$

which is $\neq \pm 1$ and provides a concrete geometric CP-odd phase.

6 Conclusion

We have shown that CP-violating phases can arise from the geometry of compact hyperbolic 3-manifolds. The holonomy of a flat $U(1)$ character around a closed geodesic, when the manifold admits an orientation-reversing isometry that fixes the geodesic's conjugacy class, yields a rephasing- invariant observable that flips sign under CP. For flat connections these phases are topological invariants, stable under metric deformations. The Meyerhoff manifold provides an explicit example realising a fifth root of unity as a CP-odd phase. We note that the twist angles $\phi(\gamma) = \operatorname{Im}\log\lambda(\gamma)$ in the A-factor of the Iwasawa decomposition [10, 11] are precisely the arguments of the flat $U(1)$ holonomies studied here; the $\mathbb{Z}/5$ torsion quantizes these phases, providing the geometric origin of CP violation in that framework. Embedding this mechanism into a complete flavor model remains open.

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Data Availability

This manuscript contains no research data.

Conflict of Interest

The author declares no conflicts of interest.

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